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Magnetostratigraphic study of the Kuche Depression, Tarim Basin, and Cenozoic uplift of the Tian Shan Range, Western China

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Abstract

Magnetostratigraphic and rock magnetic results from 1006 horizons in Paleogene and Neogene sediments between the upper Kumugeliemu Formation and the base of the Kuche Formation within the Kuche Depression of the Tarim Basin, NW China, are used to evaluate postcollisional uplift and deformation in the Tian Shan Range. Characteristic remanences with both normal and reversed polarity were isolated by thermal demagnetization, generally between 300 and 690 °C, and a composite magnetostratigraphy is constructed from data at 969 accepted levels correlating with polarity chrons C12n to C3r dated between ~31.0 and ~5.5 Ma on the CK95 geomagnetic polarity time scale. Observed height-dependent changes of rock magnetic parameters (k_m and AMS ellipsoid parameters) indicate that these sediments were influenced by weak deformation with the succession accumulated before ~15 Ma recording the effects of compressive deformation. This finding, together with two substantial increases in accumulation rate at ~16–17 and ~7 Ma, define a framework for Cenozoic uplift and deformation of the Tian Shan Range. We argue that thrusting in the south of the range was probably initiated at ~20 Ma and followed by rapid uplift at ~16–17 Ma. On the assumption that sediment accumulation in western China was still predominately tectonically controlled at ~7 Ma, another later pulse of rapid uplift with a much larger regional influence is also recognized in the Tian Shan. The Kuche Depression was subjected to two postcollisional phases of rotational deformation with clockwise rotation prior to ~12 Ma (~1.36°/Ma) succeeded by counterclockwise rotation (at ~1.64°/Ma) after ~12 Ma and possibly continuing to the present day. This rotational deformation is not temporally related to the history of basin formation resolved from the magnetostratigraphy and may be linked to block interactions with adjoining terranes.
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Keywords: Paleogene and Neogene; Western China; Central Asia; Magnetostratigraphy; Neotectonics; Magnetic fabrics; Kuche Depression; Tian Shan; Tectonic uplift

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1. Introduction

The Tian Shan Range of central Asia extends east to west for at least 2500 km and has peaks exceeding 7400 m in height. It separates the Tarim Block to the south from the Kazakhstan–Junggar "Plate" to the north (Fig. 1a) and is thought to incorporate two late Paleozoic sutures [1]. However, the remarkable deformation and elevation are predominantly related to the India–Asia collision and a protracted postcollisional (neotectonic) history has resulted from the continuing impingement of India into Asia (e.g. [2–4]). Although important for understanding Eurasian deformation induced by the India–Asia collision, the timing of neotectonic deformation in the Tian Shan Range is still unclear. The presence of a regional unconformity between Oligocene conglomerates and underlying Cretaceous strata has been used to suggest that this deformation commenced in Oligocene times [1]. However, a younger age for the bulk of the uplift and deformation is suggested by the large thicknesses of Neogene and Quaternary terrigenous sediments in the intramontane and foreland basins of the mountain range (e.g. [5,6]).

Most Cenozoic terrigenous sediments of the intramontane and foreland basins in central Asia are continental deposits and lack diagnostic fossil floras and faunas permitting precise stratigraphic assignments; volcanic rocks appropriate to radiometric age dating are also rare. In these circumstances magnetostratigraphic study becomes an important tool for constraining the history of Cenozoic uplift and deformation induced by the India–Asia collision. Magnetostratigraphy can potentially provide both direct age determination on the fossil-poor Cenozoic strata and valuable

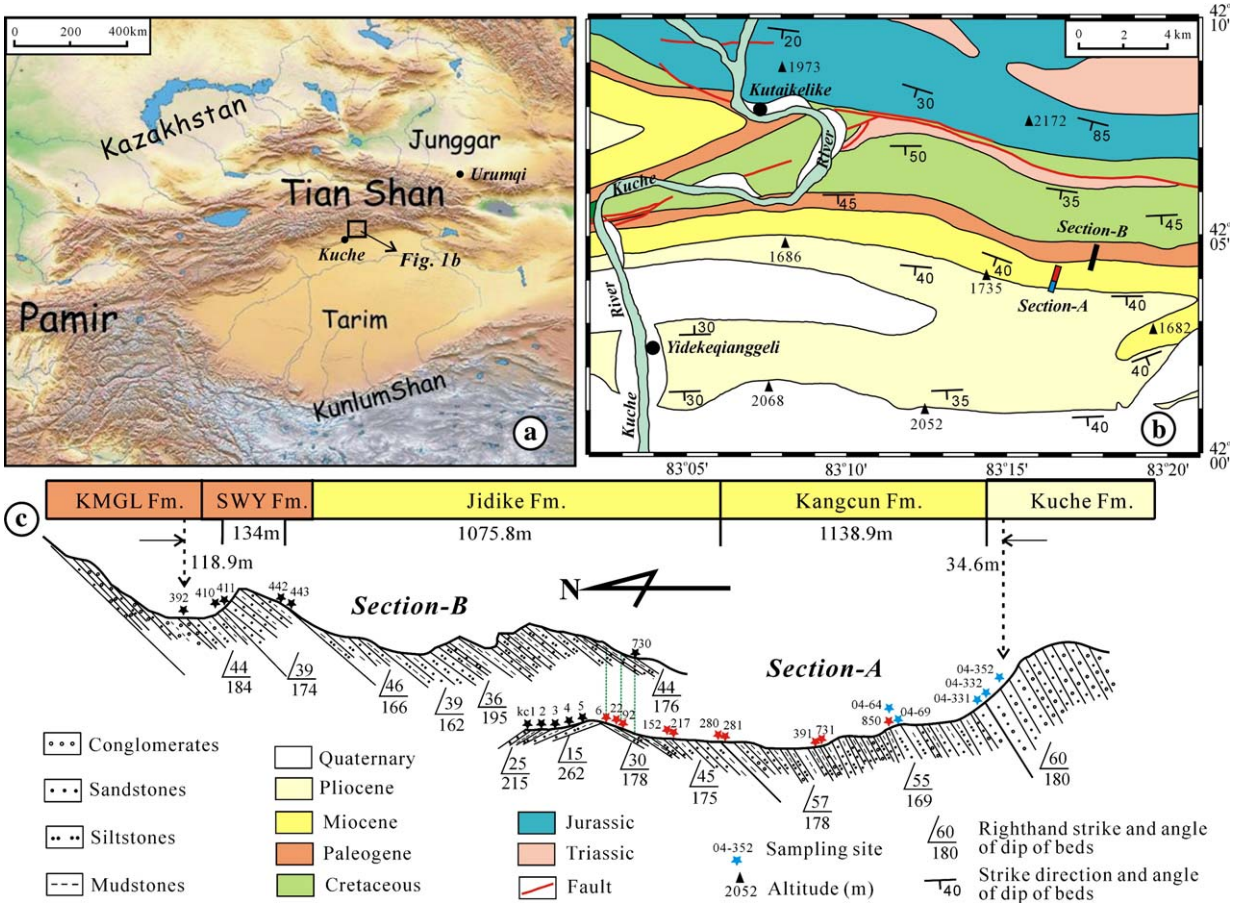


Fig. 1. (a) Topographic map of the northern Tibetan Plateau and bordering parts of central Asia showing the relationship of the study area to the Tarim and Kazakhstan blocks and bordering orogenic belts. (b) Simplified regional geologic map of the Kuche River area showing Meso-Cenozoic stratigraphy and magnetostratigraphic sampling locations of this study. (c) Composite cross section of the sedimentary succession exposed in the Kuche River area showing major stratified units and the distribution of sampling sites. The two selected sections for sampling (A and B) were lithostratigraphically correlated in the field using three marker layers of gray-green to green sandstones each with a thickness of a few meters. Abbreviations KMGL Fm. and SWY Fm. represent the Kumugeliemu and Suweiyi formations, respectively.

constraints on mountain uplift, rock denudation and climatic change.

The Cenozoic history of orogenesis in the Tian Shan Range is reviewed by Charreau et al. [17] and important facets of the geologic history are described by a number of other authors (e.g. [1,4,19–24]). Magnetostratigraphic investigations during the past decade in the foreland basins of western China (e.g. [7–18]) have aimed to link nearby orogenic uplift histories in Tibet and the Tian Shan to sediment input, and investigation in the latter region suggest that uplift of the mountain range was underway during the Miocene between 10 and 24 Ma [8,25,26]. However other studies propose that major uplift occurred much later during Pliocene and Quaternary times (e.g. [11,27,28]), a conclusion based primarily on the influx of thick conglomerates (the Xiyu Formation) in Tarim and Junggar, and within intramontane basins of the Tian Shan. In contrast, two magnetostratigraphic studies of Upper Miocene to Pliocene deposits in the foreland basins of the northern Chinese Tian Shan Range propose that major uplift was underway by ~11 and ~7 Ma [15,17], whilst investigations of mid- to late Miocene deposits in the Yaha section of the Kuche Depression at the southern flank of the mountain range [29] indicate an increase in sedimentation rate at ~11 Ma fed by orogenic uplift.

However, magnetostratigraphic studies in the vicinity of the Tian Shan Range have to date been based mostly on post–middle Miocene deposits, and no conclusions about the relationships between mountain building, rock denudation, and sediment deposition prior to mid-Miocene times have been possible. Consequently, unraveling of the full history of Cenozoic uplift and deformation in the Tian Shan requires investigation of the time interval between the Oligocene and latest Neogene and for this purpose, we have selected two Cenozoic sections from the Kuche Depression at the southern margin of the range for magnetostratigraphic and rock magnetic investigation (Fig. 1). These sections appear to cover the geologic time interval between the late Eocene/early Oligocene and the latest Miocene/earliest Pliocene. In this paper, we report this contribution to chronostratigraphic framework of Cenozoic deposition in the Kuche Depression and evaluate implications of the data for Cenozoic uplift and deformation in the Chinese Tian Shan Range.

2. Geologic framework and paleomagnetic sampling

The Kuche Depression, also known as the Kuqa or Kucha Depression, is located within the piedmont of the southern Tian Shan Range (Fig. 1a) at altitudes of ~1500 to 3000 m above sea level. This depression developed as a structurally complex foreland basin at the northern margin of the Tarim Basin as a consequence of southward thrusting of the southern Tian Shan and is recognized as one of two major Cenozoic depocenters along this margin (e.g. [5,8,22]). Approximately 10 km of terrigenous sediment accumulated in the basin as it evolved during Mesozoic and Cenozoic times of which some 5–6 km are assigned to the Cenozoic. Specifically because of its potential economic importance, the Meso-Cenozoic succession in the Kuche Depression has been the subject of detailed stratigraphic investigation and Yin et al. [8] provide a review of biostratigraphic and lithostratigraphic data from deposits of Paleogene and Neogene age. Both outcrop sequence and drill hole sections indicate that the Paleogene and Neogene succession comprises in ascending order, the Kumugeliemu, Suweiyi, Jidike, Kangcun, and Kuche formations with no major unconformities between them (e.g. [5,6,8]). This Paleogene and Neogene sequence is also generally conformable with the overlying conglomeratic Xiyu Formation [5], which is widely distributed along the margins of the Tian Shan Range and assigned a Late Pliocene to Early Pleistocene age (e.g. [9,11,15]). As observed in the Turfan and other foreland basins of the Tian Shan Range [1,19], Paleogene and Neogene successions in the Kuche Depression include a conglomerate unit at the base which overlies Cretaceous strata unconformably (e.g. [5,6]).

Two relatively complete Paleogene to Neogene sections were selected for magnetostratigraphic sampling from a site ~20 km east of a large bend of the Kuche River (Fig. 1b). The sections are separated laterally by just 2 km and thickness variations between them in formations with long strike continuity (see Fig. 1b) are small; three marker layers of distinctive gray-green to green sandstone with thicknesses of a few meters permit lithostratigraphic correlation between the two sections (Fig. 1c). As shown in Fig. 1c, section-A consists, in ascending order, of the upper Jidike, Kangcun, and lower Kuche formations, whereas section-B comprises the upper Kumugeliemu, Suweiyi, and the lower Jidike formations. The top of sampled section-A, the Kuche Formation, consists of gray-yellow to pale-yellow conglomerates with sandstone and mudstone interbeds and the lower part of section-B is generally composed of red-brown or purple-red sandstones with a few intercalated siltstones, mudstones, and gypsum beds. Two or more separate oriented drill-cores were sampled from each horizon with a sampling interval typically of 2–3 m between successive horizons although sampling intervals were sometimes as large as ~5 m where the sequence was composed mostly of conglomerate or coarse sandstone. In total, about 3000 drill-core

samples were collected from 1006 horizons with a composite thickness of ~2500 m. The lower part was sampled in the summer of 2002 and the top part of section-A (~550 m in thickness, Fig. 1c) in the summer of 2004. In addition, five sites (kc01–05) comprising 8 to 10 drill-core samples per site were collected from the northern limb of the anticline in section-A (Fig. 1c) to permit a fold test; these five sites are sandstones and siltstones correlating stratigraphically with the base of magnetostratigraphic section-A (the upper Jidike Formation). All samples were cored using a portable petrol-powered drill and oriented by magnetic compass and/or sun compass. Comparison of orientation data from the ~25% of the collection oriented by both method identified a local declination anomaly of 3° in agreement with the value estimated from the geomagnetic map of China 1990 [30].

3. Paleomagnetic results and analysis

Core samples about 2 cm in length were cut from field specimens and selected end materials were subjected to thermomagnetic analysis (magnetic susceptibility versus temperature curves) using a CS-3 apparatus coupled to a KLY-3 kappabridge. Anisotropy of magnetic susceptibility (AMS) was first measured on one specimen from each sampled horizon using a KLY-3 kappabridge. At least one specimen per horizon was then subjected to progressive thermal demagnetization in a TD-48 thermal demagnetizer. Generally, demagnetization was performed in temperature steps of 50 or 100 °C to 500 °C subsequently reduced to steps of 30, 20 or 10 °C as the maximum unblocking temperatures of the remanence carriers were approached. Magnetizations were measured by both a 2G-755R cryogenic magnetometer and a 2G-760 U-channel system and performed in the Paleomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences where magnetometers and demagnetizers are installed in a magnetic shielded house with field minimized to less than 300 nT. Demagnetization results were evaluated on stereographic projections and orthogonal (Zijderveld) diagrams [31] with the latter used to resolve component structures. Equivalent component directions were calculated by principal component analysis [32] and interval–mean results evaluated using standard Fisherian methods [33].

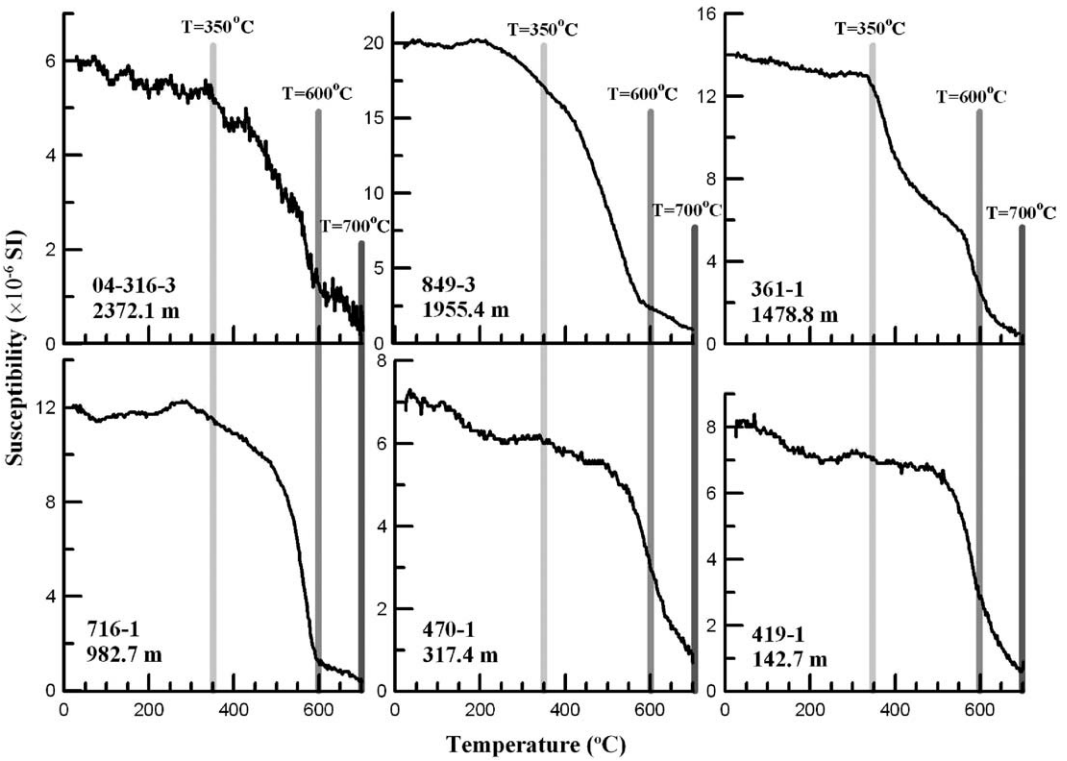


Fig. 2. Temperature dependence of magnetic susceptibility (k–T curves) for representative samples from different level of the composite section. All spectra were determined by heating from room temperature to ~700 °C in argon gas atmosphere. Results indicate that sediments from the middle and upper parts of the composite section (above ~700 m height) have relatively higher magnetite concentrations than sediments from the lower part, although all of the succession proved to be ferromagnetically dominated by hematite.

Most samples show irreversible behaviors during thermomagnetic analysis. A common characteristic in all samples is for the heating curves to show a noticeable decrease in magnetic susceptibility near the Curie temperatures of both magnetite and hematite whilst prominent increase in magnetic susceptibility generally occurs at ~580 °C in the cooling curves indicating formation of new magnetite probably from alteration of argillaceous minerals (Fig. 2). Since magnetite has a much higher specific susceptibility than hematite volume for volume, it tends to dominate the spectra when both minerals are present. The significant signature of hematite in these sediments is therefore an indication that they are dominated by hematite, with magnetite present as a subordinate ferromagnet; a few samples show significant decrease in magnetic susceptibility at 300–400 °C (sample 361-1 in Fig. 2) suggesting that maghemite or titanomagnetite is locally present as a rare ferromagnet.

Progressive thermal demagnetization trajectories are relatively straightforward in most sediments of this study and are comparable to component structures in Paleogene and Neogene sediments elsewhere in foreland basins of the Tian Shan Range (e.g. [15,17,34]). The high-temperature characteristic remanence is usually isolated following removal of a viscous recent overprint by temperature steps of 350–450 °C and clearly defined on orthogonal plots (Fig. 3). The high-temperature characteristic component (ChRM) is of dual polarity and mostly removed at temperature steps between 640 and 690 °C, or less commonly at temperature steps between 570 and 590 °C. By

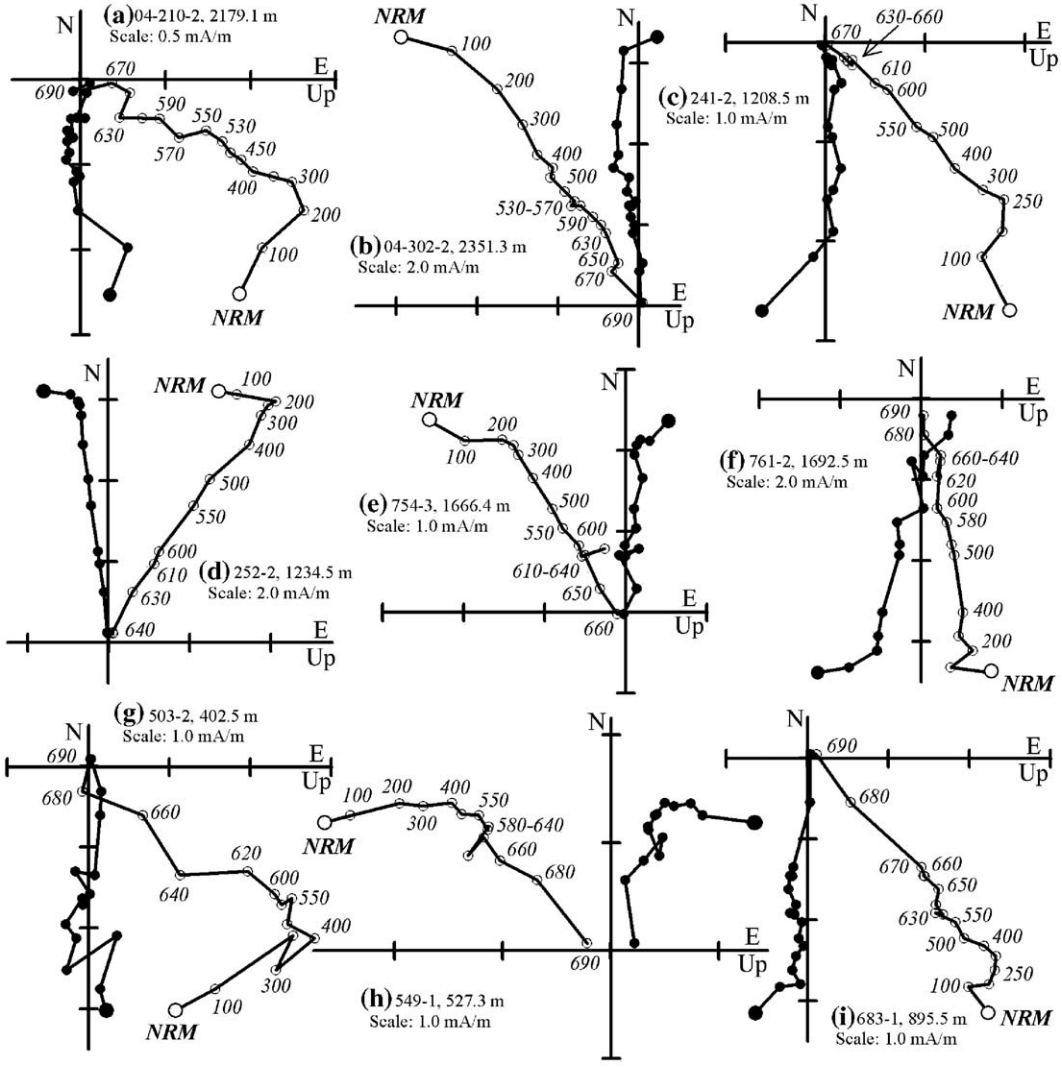


Fig. 3. Orthogonal (Zijderveld) vector plots of representative specimens from different levels in the composite section. Directions are plotted *in-situ*; solid and open circles represent vector endpoints projected onto horizontal and vertical planes respectively.

indicating that hematite and more rarely magnetite are the dominant carriers of the ChRM, the demagnetization characteristics conform to the results of thermomagnetic analysis.

Because some pilot specimens showed only unstable high-temperature demagnetization trajectories, two rounds of progressive thermal demagnetization were performed. The first stage performed on one specimen from each sampling horizon yielded stable high-temperature ChRMs in about 80% of the collection with 10 to 15% of these deviating significantly from expected Paleogene and Neogene directions in central Asia as predicted from APW paths of the adjoining major plates. Some anomalous directions have declinations trending roughly north (or south) but upward (or downward) inclinations in opposition to the predicted field and may record

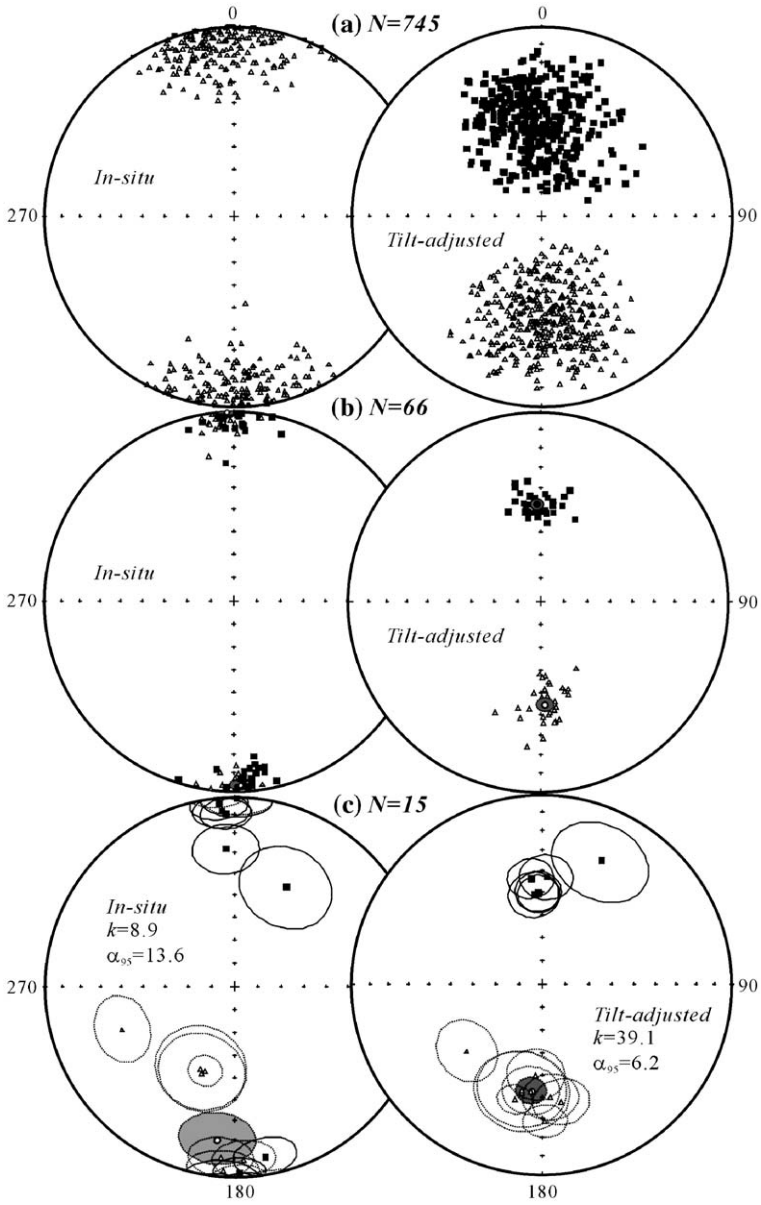


Fig. 4. (a) The 745 directions of high-temperature ChRMs comprising a coherent dipolar distribution before and after tilt adjustment; 224 anomalous directions suspected to be transitional and/or excursional geomagnetic records were excluded from these plots. (b) The stratigraphically grouped 66 interval–mean directions (Table 1) of ChRM before and after tilt adjustment. The data set defines an A-class reversal test [35]. (3) The 15 site–mean directions of ChRM from the upper part of the Jidike Formation (Table 1); these data define a positive fold test [36–38] and a C-class reversal test [35]. All plots are equal-area projections with solid/open symbols representing downward/upward inclinations.

Table 1
Summary of the interval–mean directions of ChRM from the Kuche Depression, NW China

Site ID	Height range (m)	Strike/dip	n/n0	N/R	Dg	Ig	Ds	Is	κg/κs	α_{95}g/α_{95}s
PN001	0–41.0	82/55	12/14	1/11	174.4	6.6	175.6	−48.3	16.5	11.0
PN002	44.0–82.4	82/55	10/15	1/9	172.8	18.5	173.0	−36.4	18.5	11.5
PN003[a]	85.4–115.3	98/62.5	10/14	7/3	350.1	−23.6	347.6	36.0	18.4/17.2	11.6/12.0
PN004	118.0–143.5	78/60	10/13	10/0	1.1	−12.6	6.5	45.6	23.8	10.1
PN005	145.0–180.6	108/55	15/20	13/2	358.9	−4.8	349.8	46.3	14.8	10.3
PN006	181.6–208.8	108/55	13/18	2/11	175.7	2.1	163.9	−47.4	11.3	12.9
PN007[a]	210.6–235.1	80/50	16/17	2/14	172.9	14.4	173.5	−35.7	15.1/14.8	9.8/9.9
PN008[a]	236.4–256.1	90/57	11/12	11/0	353.8	−9.6	351.0	47.1	24.1/26.7	9.5/9.0
PN009	257.8–280.8	92/58	12/15	0/12	177.0	3.8	173.5	−53.9	28.6	8.3
PN010[a]	281.9–301.9	92/58	14/15	0/14	174.8	12.3	172.0	−45.5	20.5/21.2	9.0/8.8
PN011	302.3–332.8	87/50	15/20	0/15	175.6	−4.6	174.5	−54.6	17.6	9.4
PN012[a]	335.0–366.4	88/40	14/18	14/0	354.3	−4.1	353.5	36.2	13.9/14.8	11.1/10.7
PN013	370.2–396.6	91/48	13/15	13/0	2.7	7.7	4.0	55.7	25.1	8.4
PN014	398.0–436.4	91/48	13/16	0/13	178.5	−9.8	176.4	−57.7	19.8	9.6
PN015	438.0–464.7	91/48	12/16	12/0	1.5	2.4	1.8	50.4	14.8	11.7
PN016	466.5–488.6	91/48	14/15	14/0	352.0	−6.1	349.1	41.2	17.4	9.8
PN017	489.5–513.1	91/48	12/17	0/12	176.8	7.2	175.5	−40.7	20.3	9.9
PN018[a]	514.2–537.3	92/48	9/15	0/9	171.1	−2.2	165.4	−49.3	20.1/19.5	11.8/12.0
PN101[a]	0–62.8	90/23	9/15	6/3	356.3	27.8	354.7	50.8	18.4/25.3	12.3/10.4
PN102[a]	64.6–95.3	90/31	10/11	1/9	176.9	−9.1	175.9	−39.6	22.3/20.5	10.4/10.9
PN103[a]	95.8–135.4	96/40	11/14	0/11	184.7	−10.1	184.1	−50.0	18.9/16.2	10.8/11.7
PN104	137.6–157.3	86/30	12/13	0/12	178.6	2.0	178.9	−28.1	28.7	8.2
PN105	165.3–199.7	87/40	15/16	4/11	183.7	−2.8	186.1	−42.4	12.4	11.3
PN106[a]	202.3–241.7	87/42	13/17	9/4	0.6	0.6	2.1	42.8	14.9/16.5	11.1/10.5
PN107	244.0–286.0	83/42	12/15	10/2	355.7	7.8	357.4	49.7	22.6	9.3
PN108	287.8–321.6	89/40	13/15	11/2	355.1	3.4	353.7	43.3	21.1	9.2
PN109[a]	321.6–347.9	87/41	11/12	9/2	357.1	9.5	357.4	50.4	29.6/31.2	8.5/8.3
PN110[a]	351.4–378.6	75/46	11/13	4/7	169.9	9.3	171.0	−36.7	19.7/18.3	10.5/11.0
PN111	379.3–409.9	84/49	16/17	10/6	353.5	−1.3	353.3	47.2	17.3	9.1
PN112	412.4–445.5	84/49	12/15	5/7	176.2	−2.5	177.5	−51.5	19.3	10.1
PN113	448.0–483.5	84/49	12/19	2/10	182.4	1.6	181.0	−58.4	28.2	8.3
PN114	485.5–528.6	94/60	11/16	6/5	355.0	−8.7	350.0	50.3	16.8	11.5
PN115	532.6–567.3	94/60	13/14	4/9	166.8	2.7	152.2	−57.0	29.9	7.7
PN116	570.8–610.8	93/64	9/10	1/8	169.7	12.1	162.7	−49.8	24.7	10.6
PN117[a]	616.8–670.8	95/62	9/12	3/6	172.7	9.5	165.8	−51.0	21.0/24.1	11.5/10.7
PN118	675.8–713.4	91/41	13/13	13/0	349.9	4.3	345.5	44.3	33.0	7.3
PN119	717.9–754.2	91/41	11/12	11/0	350.1	−2.3	347.1	37.9	25.7	9.2
PN120	757.2–808.3	91/41	9/14	5/4	349.1	−10.7	351.1	53.1	20.7	11.6
PN121[a]	813.0–846.3	80/57	12/13	1/11	176.7	10.7	179.6	−45.5	25.3/21.3	8.8/9.6
PN122[a]	848.8–883.0	87/57	11/14	7/4	0.3	−6.9	2.1	50.2	29.7/31.8	8.5/8.2
PN123	885.5–914.8	84/57	11/12	9/2	352.2	−6.9	351.2	50.1	29.0	8.6
PN124[a]	919.8–964.4	85/59	11/15	11/0	356.6	−5.9	357.9	53.0	28.1/26.9	8.8/9.0
PN125	971.8–1005	88/57	12/12	0/12	175.2	4.2	177.1	−50.7	22.4	9.4
PN126	1008–1037	82/55	12/13	0/12	171.9	12.2	171.8	−42.8	51.8	6.1
PN201[a]	0–144.7	96/44	10/23	6/4	2.4	6.6	0.2	50.7	12.4/13.7	14.3/13.5
PN202[a]	148.5–219.3	92/39	8/11	4/4	11.5	1.0	14.3	39.0	12.1/10.8	16.6/17.6
PN203	221.3–259.4	79/46	9/15	9/0	8.4	−2.7	14.4	39.3	13.3	14.6
PN204	259.9–295.4	75/39	11/18	9/2	3.6	10.7	12.3	46.9	9.2	15.9
PN205	301.9–346.0	108/36	10/15	3/7	191.6	−1.5	189.9	−37.2	8.8	17.2
PN206	349.2–380.1	108/36	9/12	0/9	186.0	11.5	185.2	−23.7	16.0	13.3
PN207	384.6–421.8	90/40	11/16	0/11	179.1	1.2	178.8	−38.8	20.1	10.4
PN208	425.6–467.4	85/42	9/12	3/6	197.3	1.4	202.2	−36.6	19.7	11.9
PN209	470.0–503.0	85/42	12/14	12/0	0.2	9.5	3.2	51.2	11.8	13.2
PN210	505.3–542.3	85/42	12/15	12/0	7.6	1.9	12.3	42.6	16.4	11.6
PN211	546.6–571.5	85/42	8/11	8/0	356.0	2.3	356.4	44.3	14.5	15.0
PN212	574.0–612.0	85/42	7/19	5/2	345.0	7.1	339.9	48.3	17.2	15.0
PN213[a]	614.3–669.3	94/43	8/16	6/2	15.6	9.1	22.3	51.2	16.4/14.9	14.1/14.8

Table 1 (*continued*)

Site ID	Height range (m)	Strike/dip	n/n0	N/R	Dg	Ig	Ds	Is	κg/κs	α_{95}g/α_{95}s
PN214	672.9–719.9	98/44	11/16	8/3	356.4	4.2	350.9	47.0	20.4	10.4
PN215[a]	721.9–754.4	83/47	7/13	6/1	2.3	0.8	6.7	47.4	11.1/10.5	19.0/19.6
PN216	757.1–788.4	86/43	13/15	13/0	1.1	1.6	3.5	44.2	20.3	9.4
PN217	792.4–828.2	86/43	12/14	3/9	187.1	−0.4	191.8	−41.9	18.1	10.5
PN218	829.5–861.8	83/43	10/13	2/8	177.7	6.9	178.7	−35.9	22.5	10.4
PN219[a]	863.9–900.0	84/43	10/13	7/3	2.9	8.0	7.8	50.7	13.6/14.1	13.6/13.3
PN220	903.0–939.8	89/44	11/13	1/10	174.5	2.5	173.1	−41.4	26.7	9.0
PN221[a]	941.8–977.7	90/43	9/14	4/5	179.2	−14.2	178.7	−56.9	53.3/51.0	7.1/7.3
PN222	982.7–1027	91/41	12/14	2/10	179.6	−6.9	179.0	−47.9	22.1	9.4
Mean			34/66	34/0	358.2	0.6	358.4	47.0	53.1/94.6	3.4/2.5
			32/66	0/32	177.7	2.8	176.9	−45.2	63.2/56.7	3.2/3.4
			66/66	34/32	358.0	−1.1	357.7	46.1	57.0/71.7	2.3/2.1
KC01[b]		218/25	8/8	0/8	249.1	−36.4	228.3	−45.5	18.9	13.1
KC02[b]		238/15	5/6	0/5	202.5	−51.0	190.8	−41.0	17.5	18.8
KC03[b]		232/12	5/8	0/5	201.0	−49.2	191.0	−42.1	19.7	17.7
KC04[b]		262/15	8/8	0/8	199.3	−50.9	193.3	−37.2	66.2	6.9
KC05[b]		283/12	5/8	5/0	27.3	40.3	25.4	28.6	17.7	18.7
Mean[c]			15/15	6/9	6.5	19.9	6.0	42.8	8.9/39.1	13.6/6.2

Abbreviations are: Site ID, site identification; height range, relevant height range for grouped site in each section as described in Figs. 5 and 6; Strike/dip, strike azimuth and dip of bed; n/n_0, number of samples or sites used to calculation/yielded well-defined ChRM or demagnetized; N/R, samples or sites show normal/reversed polarity; D_g and I_g, κ, α_{95} (D_s and I_s, κ, α_{95}), declination and inclination of direction, precision parameter, 95% confidence limit of Fisher statistics *in-situ* (after tilt adjustment).

[a] The bedding attitudes are given by averaging values.

[b] Sites were collected from the northern limb of anticline in section-A as described in Fig. 1c.

[c] Mean direction for sites KC01–05 and PN101–110 from the upper part of the Jidike Formation.

transitional or excursional geomagnetic fields. To optimize recovery of useful information a second round of demagnetization was performed on horizons that initially showed erratic or anomalous high-temperature trajectories. Following demagnetization 969 out of 1006 sampled horizons yielded well-defined ChRMs with 745 component directions clustering within 40° of the mean in a NW-to-NE quadrant with moderate positive inclinations, and the equivalent antipodes (Fig. 4a). These ChRM directions are grouped stratigraphically into 66 sites (Fig. 4b) each one with an average thickness of ~30–40 m and typically incorporating ~10 components (Table 1). The overall group–mean direction is D=358.0°, I=−1.1°, κ=56.8, α_{95}=2.3° before tilt adjustment and D=357.7°, I=46.1°, κ=71.5, α_{95}=2.1° after tilt adjustment.

Since structural dip is fairly uniform across the region the fold test is inconclusive although grouping of the interval–mean directions is marginally improved following tilt adjustment. A reversal test [35] however, is positive with an angular difference of 1.1° between tilt-adjusted interval–mean directions of each polarity (Fig. 4b). This is less than the critical angle of 4.2° and yields an A classification reversal test. In addition, 31 of 38 demagnetized specimens from sites KC01–05 (Fig. 1c) with contrasting tectonic orientation and sampled specifically for the purpose of applying a fold test, also yield well-defined high-temperature ChRMs. Details for these five sites and the stratigraphically grouped 66 sites are listed in Table 1. The sites kc01–05 are supplemented by sites PN101–110 from the upper part of the Jidike Formation to yield a 15 site formation–mean direction that is significantly improved by tilt adjustment with κ_s/κ_g=4.4 (Fig. 4c) and defines a positive fold test [36,37]. Further analysis using the Watson and Enkin [38] test presents an optimal concentration at 102.2% unfolding with the 95% uncertainties ranging from 90.2 to 112.7% unfolding and strongly indicating that this remanence is pre-folding. The reversal test [35] is also positive for these 15 site–mean directions (γ_0=5.2°<γ_c=12.9°, C classification). Although the age of regional folding in the Kuche Depression might be younger than Pliocene, the collective reversal and fold tests indicate that the ChRM was acquired at, or close to, the time of rock formation and is compatible with a primary remanence of post depositional detrital origin (PDRM).

4. Magnetostratigraphy

The directions of ChRM from 969 horizons were used to calculate virtual geomagnetic pole (VGP) latitudes and the sections organized into stratigraphic levels referred to the VGP latitudes (Fig. 5) to define magnetic polarity zones. The two studied sections are separated by about 2 km and were correlated in the field using lithostratigraphic criteria comprising three marker layers of distinctively colored sandstone. This correlation identifies an overlap of ~100 m (Fig. 1b and c) and when the collective magnetic polarities, magnetic susceptibility, and anisotropy of magnetic susceptibility (AMS) records from the overlapping segments are compared this estimate is refined to identify an actual overlap of approximately 120 m (Figs. 5 and 6). Excluding magnetic polarity zones suggested from one horizon only, a total of 41 pairs of normal (N1–N41) and reversed (R1–R41) polarity zones are clearly identified in the composite section (Fig. 5).

The magnetic polarity sequence can be satisfactorily correlated with the CK95 geomagnetic polarity time scale (GPTS) [39] because certain levels in the Paleogene and Neogene successions of the Kuche Depression yield fossils of foraminifera, bivalves, algae, ostracoda and gastropods in both field and drill-core sections that can be used to assign biostratigraphic ages ([8] and references therein). The oldest (Kumugeliemu) formation is considered to be late Paleocene to early Miocene in age; the gypsum-bearing Suweiyi Formation ranges from Eocene to early Miocene, and the Jidike and Kangcun formations above are most likely to range from Oligocene to Miocene. The top unit of the studied sections, the Kuche Formation, is dominated by coarse-grained sandstones and conglomerates with minor thinly bedded mudstones and siltstones and has poor biostratigraphic constraint (Fig. 1c and [8]) but is roughly conformable with underlying formations, and is overlain by the conglomeratic Xiyu Formation. It may also correlate with the Atushi Formation in the Kashgar region, a Cenozoic depocenter of the Tarim Basin comprising another foreland basin to the Tian Shan Range (e.g. [5,6]). Magnetostratigraphic age estimates of the Xiyu Formation mostly range from ~1.8 Ma to ~3.5 Ma [9,11,15] and preliminary results indicate that the Atushi and Kuche formations are dominated by a long reversed polarity zone believed to correlate with the Gilbert reversed chron [9,40]. We therefore suspect that the succession studied here terminates in late Miocene or Early Pliocene times at ~5 Ma.

The completeness of the composite section, the apparent absence of major stratigraphic breaks, and the spacing of observed magnetic polarity intervals all suggest that most polarity events during the latter part of the Cenozoic are represented here. As shown in Fig. 5, local magnetozones (R11–R26) with a distinctive interval pattern in the lower part of section-A and the top of section-B (at ~800–1600 m height in the composite section of Fig. 6) are dominated by reversed polarity with a zone punctuated by mixed polarity and short events (N12–N21) interbedded between two relatively thick reversed polarity zones (R11 to R12 and R22). This pattern of magnetozones appears to correlate with chrons C5n.2r to C5Dr of the CK95 GPTS [39]. In the upper part of section-A (i.e. at ~1600–2400 m height in the composite section, Fig. 6), the distinctive and well-established magnetozones (N1–N10) then appear to correlate with chrons C3An.1n to C5n.2n of the CK95 GPTS. The distinctive magnetozones between ~600 and 800 m height in section-B (N26–R31) are characterized by two relatively thick normal polarity zones (N26 and N27) preceded by three pairs of normal and reversed polarity zones readily correlated with chrons C5En to C6AAr of the GPTS. Although there is no clear match to thick zone N31, the magnetozones N32–N41 in the lower part of section-B can be correlated with most of the chrons between C6Bn and C12n, and N37 to N41 correlate well with polarity chrons between 26 and 30 Ma. Hence the number of reversals, and in large measure the relative thicknesses of the polarity zones, resolved in this study suggest that the Kuche Depression succession was deposited between ~31 Ma and ~5.5 Ma (early Oligocene to latest Miocene) and that the Paleogene/Neogene boundary is located within the lower part of the Jidike Formation.

A plot of magnetostratigraphic age versus height within the composite section (Fig. 7) shows two abrupt increases in the rate of sediment accumulation from ~7 to ~13 cm/ky at ~16–17 and from ~13 to ~23 cm/ky at ~7 Ma. These two changes in sedimentation rate correlate approximately with the two main pulses of sedimentation in the Tian Shan region at ~17 and ~5–6 Ma estimated from the entirely independent method of balanced section reconstruction [22]. They also conform to the estimate of Avouac et al. [20] of 16 +22/−9 Ma for initiation of deformation in the Tian Shan. In addition, the computed sedimentation rates ranging from ~7 cm/ky during Oligocene to mid-Miocene times, to ~13 cm/ky by the mid- to upper Miocene and to ~23 cm/ky in the latest Miocene (Fig. 7) are comparable with rates estimated elsewhere in the Northern Tibetan Plateau, in the Tian Shan Range, and on both margins of the Himalaya (e.g. [10,11,16,17,41]); they also appear to be typical of sedimentation rates in foreland-basin environments (10–40 cm/ky, [42,43]).

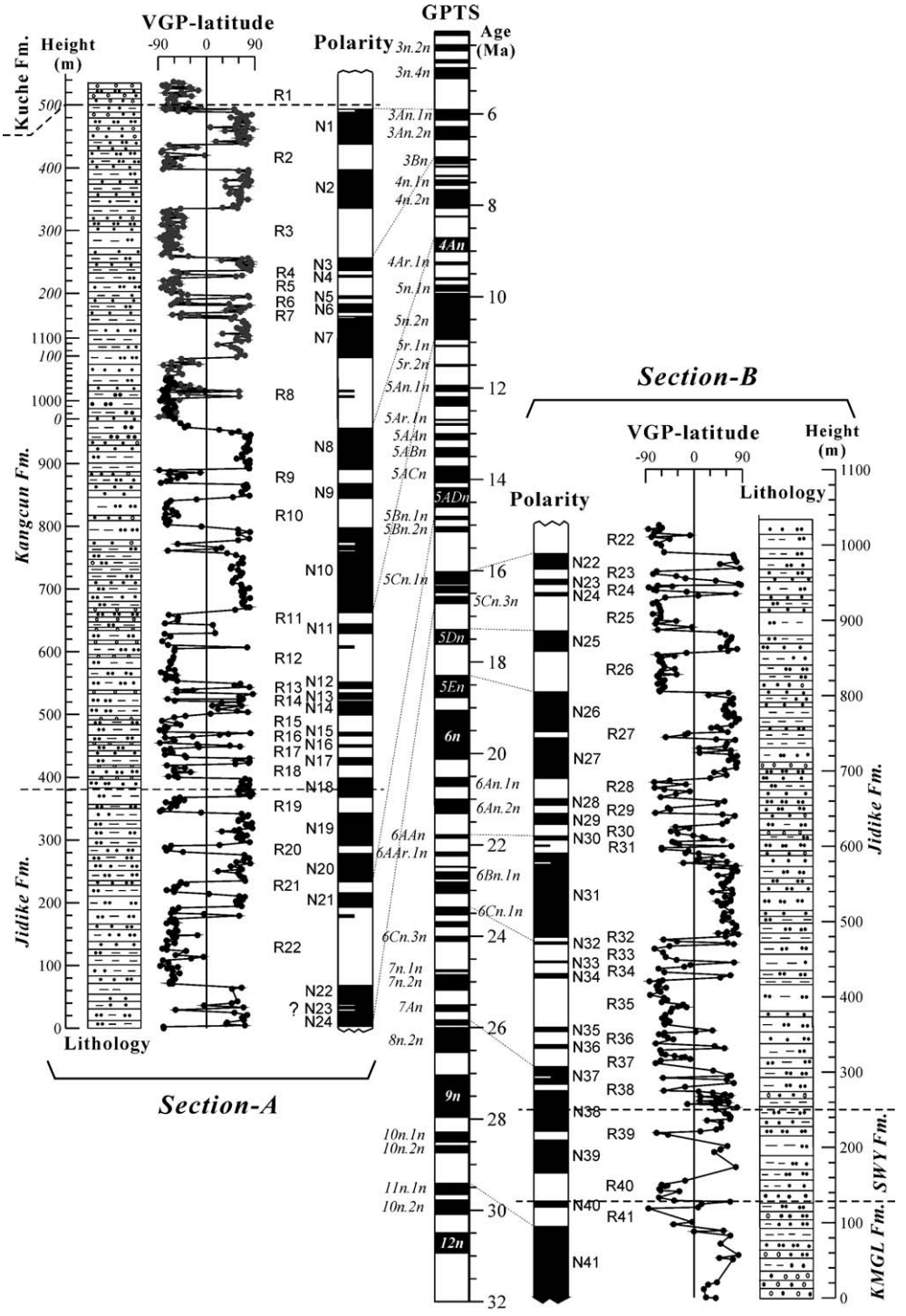


Fig. 5. Lithology and magnetostratigraphic results from the two sections in the Kuche Depression, NW China with VGP latitude plotted as a function of stratigraphic level and showing the correlation with the CK95 GPTS [39] described in the text. Magnetochrons defined by only one horizon are shown by short-half bars. Lithologic legends are the same as for Fig. 1.

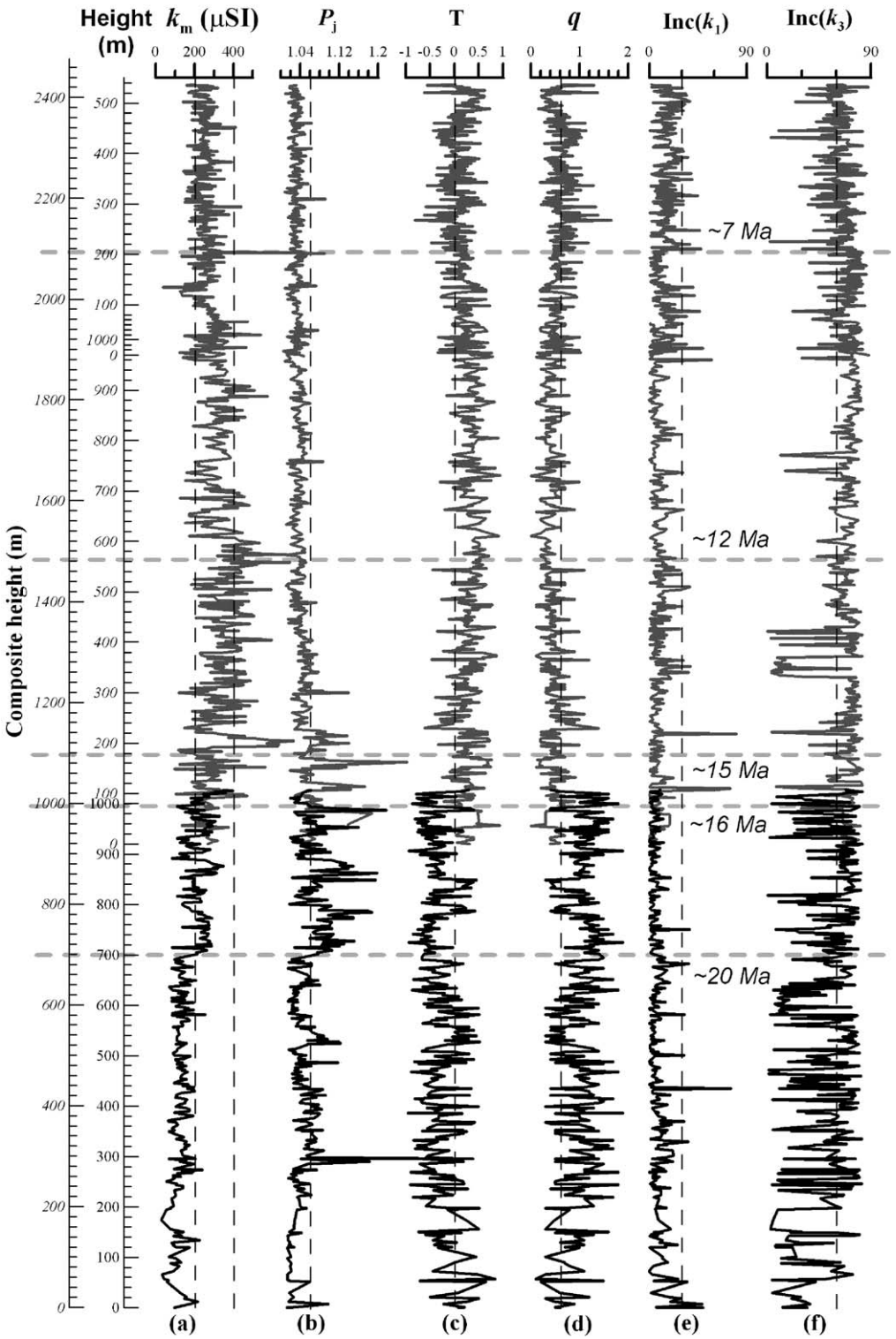


Fig. 6. Rock magnetic results from the composite section in the Kuche Depression, NW China, comprising mean magnetic susceptibility (k_m), corrected AMS degree (P_j), AMS shape parameter (T), and inclinations of the maximum (k_1) and minimum (k_3) principal axes of the AMS ellipsoids all shown as a function of height. The heights on the left side of the diagram show the stratigraphic height above the base of the composite section.

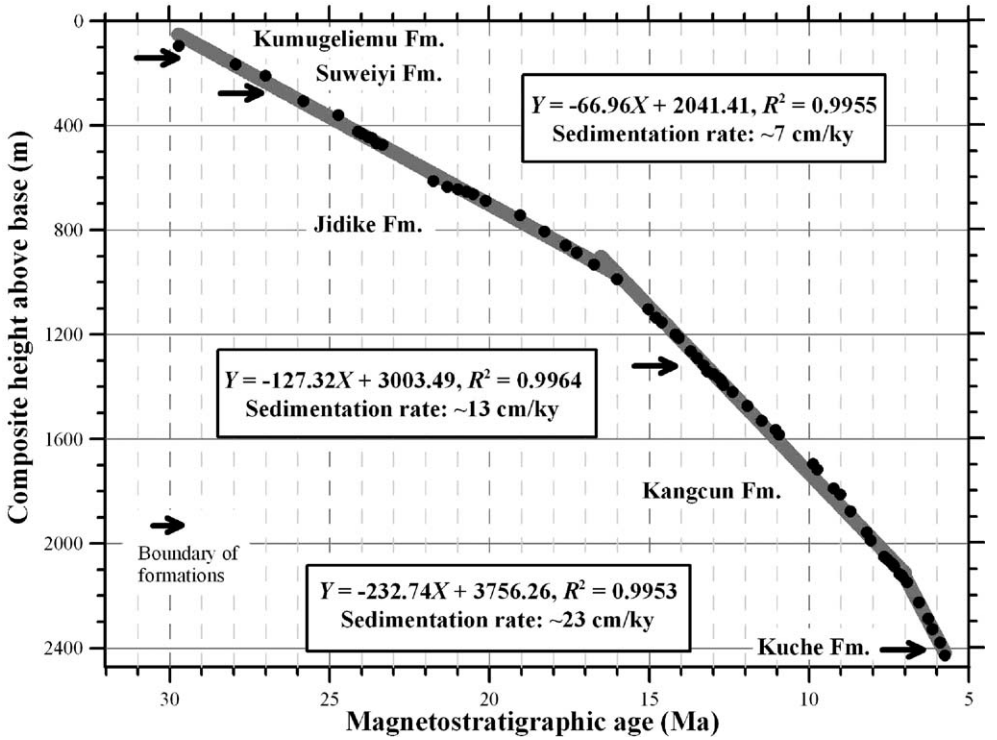


Fig. 7. Stratigraphic height as a function of magnetostratigraphic age in the composite stratigraphic succession from the Kuche Depression. Two distinct increases in accumulation rate are identified at ~16–17 and ~7 Ma respectively. All age constraints are determined by correlating the polarity chrons with the CK95 GPTS [39] as described in the text and shown in Fig. 5.

A recent magnetostratigraphic study by Charreau et al. [29] of the Yaha section to the southeast, and on an approximate strike continuation, of our study covers a shorter time interval recording 18 normal and 17 reversed chrons (~1000 cores with 2 cores per horizon and an average site separation of 5.6 m). These authors infer a doubling of the sedimentation rate at ~11 Ma and infer that this was when uplift and erosion of the Tian Shan accelerated. However, examination of their magnetostratigraphic correlation shows an equally valid correlation with the GPTS [39] between ~2 (C2r.1n) and ~10 Ma (C5n.1n) suggesting that the Yaha succession incorporates the acknowledged higher rates of deposition that have characterized Pliocene and Pleistocene times and overlaps only with the youngest part of our succession. This alternative correlation suggests that the increase in sedimentation rate observed by these authors actually occurred at ~7–8 Ma, which is where we observed it in our section, and the estimated sedimentation rate in the lower part of the Yaha succession (~20 cm/ky) is then closer to the sedimentation rate (~13 cm/ky) found in the higher part of the succession studied here prior to ~7 Ma. This alternative correlation would also lower sedimentation rates in the younger part of the Yaha section to ~32 cm/ky which is closer to the rate observed within the last ~1.5 Ma of our succession (Fig. 7) and more compatible with sedimentation rates of 10–40 cm/ky observed in foreland-basin environments elsewhere [42,43].

5. Anisotropy of magnetic susceptibility (AMS)

Magnetic susceptibilities and AMS fabrics were determined for 976 specimens from the 1006 sampled horizons and show significant changes with height in the composite section (Figs. 6 and 8). Specifically, mean magnetic susceptibility (k_m) is lowest (<200 μSI) near the base of the succession; it displays a marked increase at ~700 m height within the composite section at an estimated age of ~20 Ma (Fig. 6a) and then progressively increases to reach a maximum (>400 μSI) at ~1500 m above the base of the succession at a level estimated to be ~12 Ma in age. It remains at a somewhat lower value to ~2000 m above the base of the succession at which point it falls further to a value mostly in the range of ~200–300 μSI until the top of the composite section is reached at ~2400 m. The overall relatively weak values of bulk susceptibility (mostly <500 μSI) suggest that anisotropic paramagnetic and

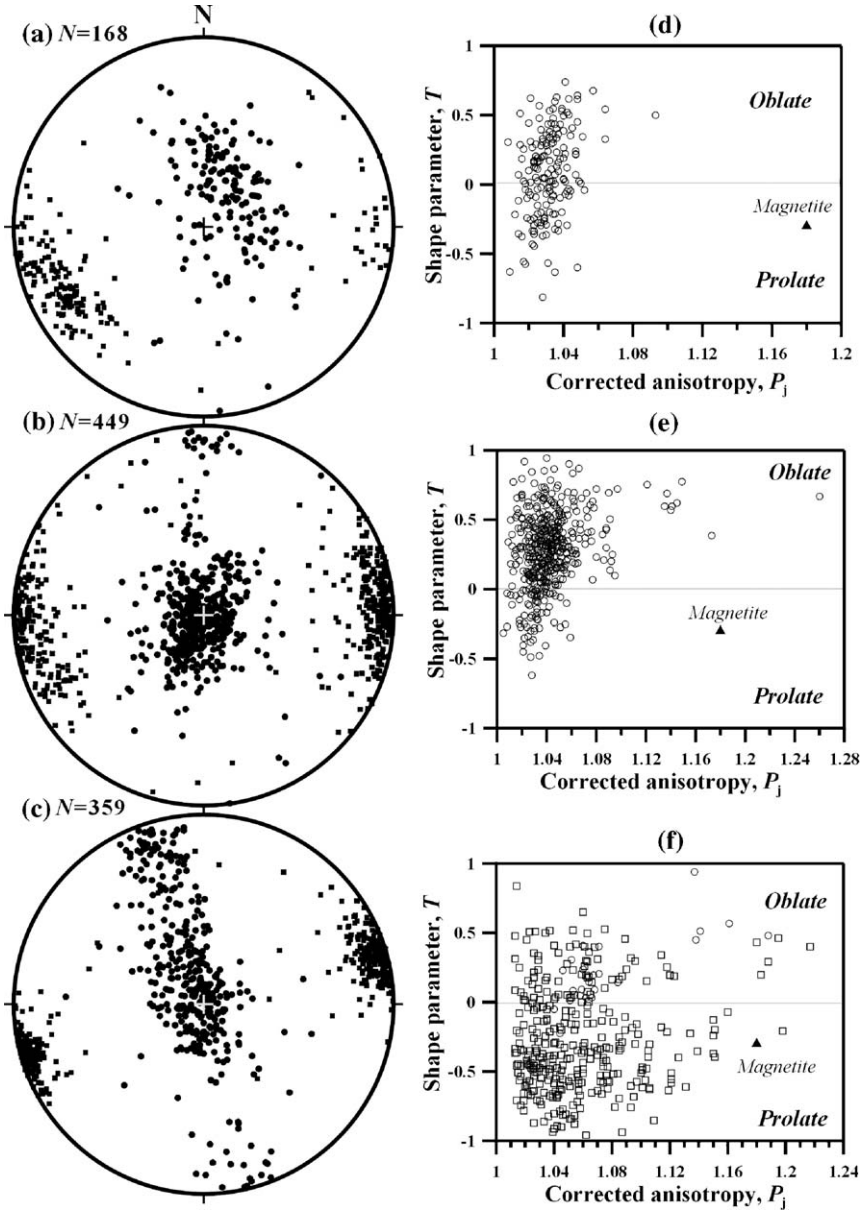


Fig. 8. Stereographic projections and P_j–T (corrected AMS degree versus AMS shape parameter) diagrams for AMS data from the Kuche Depression, NW China. To illustrate critical changes in these properties through the succession the AMS data are subdivided into results (a, d) above ~2000 m, (b, e) between ~1000–2000 m, and (c, f) from the base to ~1000 m height above the base of the composite section. The AMS ellipsoid for magnetite (P_j=1.18, T=−0.30, [44]) is shown by the triangle in (d), (e) and (f). All directions shown on the stereoplots are in the lower hemisphere.

antiferromagnetic minerals such as the clays and hematite respectively are the dominant control of both susceptibility and anisotropy in these sediments [44] although the noticeable increase of k_m above ~700 m and especially at 1100 m could indicate increased input of magnetite after ~20 Ma. Magnetic properties of sediments are typically dominated by magnetite when present as ≥0.5% of the iron oxide fraction [44] and both rock magnetic and thermal demagnetization behaviors (Figs. 2 and 3) indicate a more significant temperature dependence of magnetic susceptibility around 580 °C in the middle and upper parts of the composite section (Fig. 2). The duration and consistency of the susceptibility signature implies that this is a tectonic signature of enhanced magnetite-rich detrital influx rather than the result of climatically-controlled pedogenic processes [10,16].

The most notable changes in corrected anisotropy degree (P_j) and shape parameter (T) [45] take place at the

~1000–1100 m level in the composite section with an assigned magnetostratigraphic age of ~15–16 Ma (Fig. 6b–c). Changes in orientation of the principal anisotropy axes also occur at this level (Figs. 6e and 8a–c). The height-dependence of AMS character does not correlate specifically with changes in k_m and is apparently independent of the ferromagnet (*sensu lato*) concentration in the succession as is also observed in the Subei area of the Qaidam Basin [10]. Furthermore the elongated distribution of AMS ellipsoids (P_j–T data) does not pass through, or towards, the magnetite ellipsoid (Fig. 8d–f) and indicates that the observed AMS character is not a straightforward function of magnetite concentration changes in the succession [46].

From the base to the ~1000–1100 m level in the composite section, sample values oscillate about relatively large P_j values with marked enhancement of P_j at the ~700–1100 m level, while samples above the ~1100 m level show low P_j values mostly less than 1.05 (Fig. 6b). Correspondingly, the ellipsoid shapes are dominantly prolate ($T<0$) from the base to the ~1000 m level and then change to predominantly oblate ($T>0$) in the middle and upper parts of the composite section (Figs. 6c and 8d–f). This is also true for the orientation of the minimum, k_3, principal anisotropy axes (Figs. 6e and 8a–c) which are largely distributed within a near N–S girdle to ~1000 m above the base of the succession and then change to an orientation perpendicular to bedding above the ~1000 m level. In contrast directions of maximum susceptibility axes (k_1) show little change with height above the base of the succession and remain nearly horizontal with E–W to ENE–WSW trends throughout (Figs. 6d and 8a–c). This k_1 trend is approximately parallel to fold axes in the Cenozoic strata within the Kuche Depression (e.g. [5,8,17]). In addition, a slight change in AMS ellipsoids is observed at the ~2000 m level within the composite section (Figs. 6c and 8) and above this level the sediments tend to show weak and neutral AMS shapes and k_3 directions inclined to the north and northeast (Figs. 6c, 8a, d). Since sediment input has been predominantly from the uplifting Tian Shan to the north, we might expect the k_1 axes to be distributed predominantly about a N–S direction if the fabric was of primary depositional origin (e.g. [44,47]). Instead the uniform focused E–W trend actually observed and the height-dependent nature of the AMS fabric is comparable to embryonic magnetic fabrics in weakly deformed mudrocks (e.g. [46,48,49]) and implies that the rocks in this succession have been subjected to incipient deformation.

The distinct transition of AMS ellipsoids from prolate-dominated in the sedimentary succession from the base to the ~1100 m level and then to oblate or neutral shapes above this level is broadly comparable to a transition observed in the sedimentary succession from the Subei region of the Gansu Province of Western China [10]. However, the observed height-dependent changes in the succession studied here does not correspond to a transition in sedimentary environments from relatively far traveled fine grains to poorly rounded grains transported a short distance [10]. Rocks from the base to the ~1100 m level are dominated by prolate rather than neutral shapes (Fig. 8f) and k_1 axes are tightly grouped in the lower part of the succession and become more dispersed at higher stratigraphic levels (Fig. 8a–c). Several studies have shown that magnetic fabrics in mudrocks can undergo a series of changes during weak deformation (e.g. [46–51]) in which distinct changes in magnetic ellipsoids are characterized by a cluster of k_1 directions perpendicular to shortening direction, with k_3 directions distributed from normal to bedding to a clear girdle, and magnetic ellipsoids grading from the oblate field (sedimentary fabric) towards the prolate field. Furthermore, Parés [46] defined a general P_j–T path to show the development of AMS ellipsoids as the intensity of deformation increases in weakly deformed rocks. Thus the observed height-dependence of AMS fabrics seems to incorporate a signature of tectonic strain which imparted an embryonic, but magnetically detectable, signature in this succession.

The Tibetan Plateau and its northern hinterland have been subjected to N–S directed compression since onset of the India–Asia collision and significant crustal shortening and lithosphere thickening in this vast region have been resolved from a range of studies ([52] and references therein). Because weak strains commonly cause k_1 directions to become perpendicular to σ_1 (e.g. [46–51]), we suspect that the dominant k_1 directions observed in the lower part of this succession (Figs. 6d and 8c) are a response to regional compression induced by the India–Asia collision. The observed height-dependent changes of AMS ellipsoids, P_j–T data (Fig. 8d–f) and the distribution of k_3 (and k_2) directions (Figs. 6e and 8a–c), can also be related to a general P_j–T path for the development of AMS ellipsoids in weakly deformed rocks with increasing deformation intensity as shown by Parés [46]. The dominant prolate shapes and nearly N–S girdle of the k_3 directions in rocks from the base of the succession to ~1100 m (Figs. 6c, 8c, f) are in the older and more deeply buried part of the succession presumably exposed to more intense strains than the succession above the ~1100 m level. At higher levels AMS ellipsoids mostly have oblate shapes and the k_3 directions are distributed roughly normal to bedding (Figs. 6c, 8b, e) whilst above the ~2000 m level AMS shapes and k_3

directions show evidence for southward imbrication of possible primary sedimentary origin (Fig. 8a, d, [10]).

In general terms the observed height-dependent changes of AMS ellipsoids indicate that a tectonic strain field induced by the Cenozoic India–Asia collision existed prior to ~15 Ma, whereas the signature of straining is much weaker or absent in the younger and less deeply buried part of the succession (c.f. Figs. 6 and 8). The primary nature of the remanence in the lower parts of the succession is established from the consistent sequence of polarity zones and the paleomagnetic field tests. It therefore appears that magnetic grains fixed by PDRM processes have not been substantially reoriented by the weak regional compression and this supports an early origin for the embryonic magnetic fabric in weakly deformed mudrocks ([49] and references therein).

6. Discussion

6.1. Cenozoic uplift of the Chinese Tian Shan Range

The Paleogene and Neogene Kuche Depression is a foreland basin formed by flexural subsidence resulting from southward thrusting and nappe stacking near the southern rim of the Chinese Tian Shan Range (e.g. [5,6,53]). The Paleogene and Neogene sedimentary infill is a direct response to thrusting and loading of the crust during deformation and provide a unique signature of orogeny and erosional history. As observed in the Turfan Basin and some other intramontane and foreland basins of the Tian Shan Range [1,19], the Kumugeliemu Formation at the base of the Paleogene comprises a laterally extensive unit of conglomerates and sandstones succeeded by gypsum-bearing sandstones and mudstones (e.g. [5,8,53]). This succession is seen to lie on underlying Cretaceous strata with distinct unconformity (e.g. [5,6,53]). This succession has previously been interpreted as the onset of Cenozoic thrusting within the Tian Shan Range following uplift and unconformity formation [1,19] or alternatively as onset of sedimentation following eustatic changes during Early Oligocene times [22]; the latter view excludes a specific link to the Tian Shan and is based primarily on the wide extent of this unconformity throughout the Kunlun Shan and Tian Shan ranges with these events embracing a long-term fall in sea level culminating in a short term fall at ~30 Ma [54]. The base of section-B studied here (Fig. 1c) starts mostly above the conglomerates of the Kumugeliemu Formation with the magnetostratigraphic correlation consistent with top of this formation being ~29 Ma (mid–late Oligocene) in age (Fig. 5); this formation is attributed to a shallow marine or shoreline origin [22].

The boundary of the Kumugeliemu and Suweiyi formations in the Kuche Depression is interpreted as a rapid transition from marginal-marine and lacustrine to fluvial and alluvial facies [8]. The magnetostratigraphic evidence indicates a maximum age of ~29 Ma for this contact in our section. Yin et al. [8] suggest that this marine to terrestrial facies transition was a consequence of the onset of thrusting in the Chinese Tian Shan. However, recent reconnaissance [53] and our field investigations show that local marine transgression and saline deposition are also observable near the base of the Jidike Formation and imply that oscillations in marine and terrestrial environments continued at least until ~26 Ma in this region and cannot therefore be directly related to the onset of thrusting.

Rock magnetic parameters (k_m and AMS ellipsoid shape) show notable changes within the lower part of the Jidike Formation at a height of ~700 m in the composite section with an estimated age of ~20 Ma (Figs. 6 and 8) and marked increase in k_m suggesting influx of magnetite-rich detritus. Height-dependent changes in AMS ellipsoids occur in sediments dated ~15 Ma with older deposits recording the influence of compressive stress; from this evidence we infer that thrust initiation in the southern Chinese Tian Shan actually occurred at about 20 Ma. Specifically this follows from: (i) a contrasting rock provenance as indicated by the abrupt increase in k_m (Fig. 6a); (ii) a sharp increase in the parameter P_j (Fig. 6b), and (iii) a near shore to lacustrine to fluvial transition in sedimentary facies within the lower part of the Jidike Formation (Figs. 1c, 5 and [53]). The proposed initiation of thrusting in the southern Tian Shan Range at ~20 Ma is compatible with the estimated commencement of thrusting in the Subei section of the northern Tibetan Plateau [10] and with early Miocene unroofing ages determined by apatite fission track dating in the northern Tibetan Plateau and Tian Shan Range [4,55,56].

Rapid uplift of the southern Chinese Tian Shan Range appears to have been initiated at ~16–17 Ma, a few million years later than the onset of thrusting. As shown in Fig. 7, sediment thickness in the Kuche Depression indicates an abrupt increase in sedimentation rate from ~7 to ~13 cm/ky at this time and this correlates with increase in conglomeratic beds in the higher part of Jidike Formation (Fig. 5 and [53]). Whilst it is acknowledged that regional and local climatic changes may also contribute to increase in erosion and accompanying accumulation rates, this is considered to be a subsidiary, and possibly insignificant, contributor to magnetic signatures observed here because Oligocene and Miocene climatic conditions in western China appear to have been relatively uniform (e.g. [57,58]) and they are unable to explain the observed rapid changes

followed by long intervals of uniform magnetic properties (Figs. 5 and 7). The presence of conglomeratic interlayers in the mid-Miocene to Pliocene sequence within the Kuche Depression (Figs. 1c, 5 and [8,53]), as well as elsewhere in the northern Tibetan Plateau and central Asia (e.g. [15–17]), indicate that observed increases in sediment accumulation rates predominantly reflect mountain uplift. Thus, the relatively constant sediment accumulation rate between ~16–17 Ma and ~7 Ma (Fig. 7) suggests that a balance between uplift, erosion and deposition prevailed in the southern Chinese Tian Shan during this period.

The other notable increase in sediment accumulation rate in the Kuche Depression at ~7 Ma (Fig. 7) may be the signature of a pulse of rapid uplift comparable to that observed in the northern Chinese Tian Shan Range [15]. In the Kuche Depression, this rapid uplift pulse correlates with changes in rock magnetic parameters and AMS ellipsoid form and may be the signature of a further tectonically-controlled change in sediment provenance accompanying reactivation of the southward imbrication and nappe stacking along the southern margin of the Tian Shan. The two pulses of rapid uplift in the southern Chinese Tian Shan at ~16–17 and ~7 Ma conform to the main pulses of sedimentation at ~17 Ma and ~5–6 Ma in the Tarim and Junggar basins estimated from balanced section reconstructions [22]. The relative contributions of tectonic uplift and climatic changes to the subsequent increases in sedimentation rates observed in the Pliocene and Quaternary successions is still unresolved (see [15,59–61]).

6.2. Postcollisional rotation patterns of the Kuche Depression

The stratigraphically grouped 66 interval–mean ChRM directions (Table 1) identify the history of Cenozoic rotation of the tectonic block incorporating the Kuche Depression. The average interval–mean declination yielded by these sites (D=357.7±3°) is close to the declination recorded in the Yaha section (D=353.5±3°, [29]) and marginally anticlockwise from both the mean dipolar magnetic field and present geomagnetic field (D=~3°). Comparison between the declination record in the Kuche depression and the expected regional declination computed from the master APW path of Eurasia at 30, 20, and 10 Ma [62] identifies a progressive counterclockwise (CCW) rotation of the Kuche Depression by ~10° relative to Eurasia (from 13.1±5.4° at 30 Ma to 8.9±3.6° at 10 Ma). Subsequently the region was subjected to a CCW rotation of 7.8±7.1° relative to Tarim. A mean Paleocene to Miocene paleopole computed from four studies derived from three sections at the southern margin of the Tarim Basin

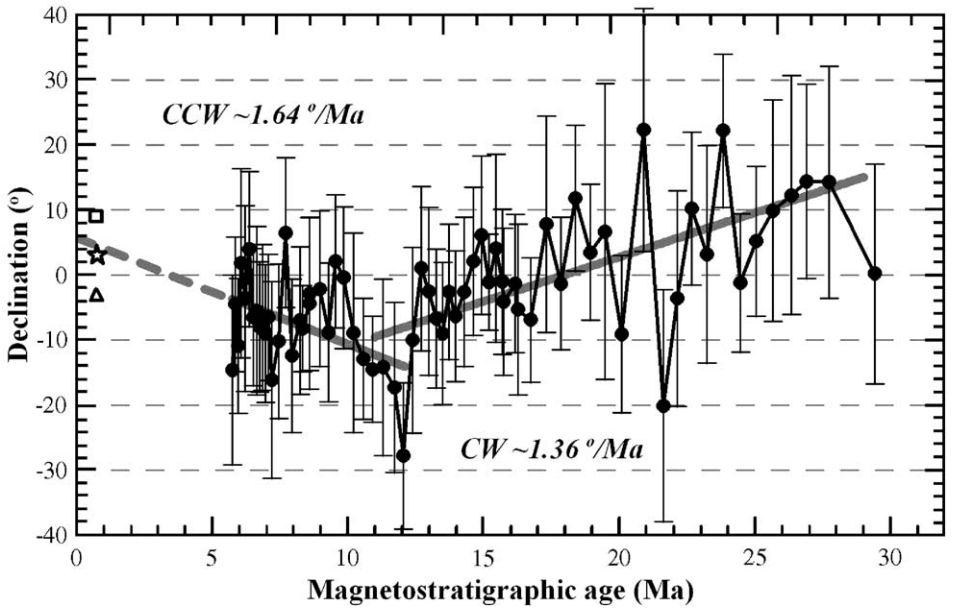


Fig. 9. Plot of declination of the stratigraphically grouped 66 interval–mean ChRM directions (Table 1) versus magnetostratigraphic age in the composite section within the Kuche Depression, NW China. From these data two episodes of rotation relative to the geographic frame are interpreted to have influenced the block incorporating the Kuche Depression during the last ~30 Ma (fitted lines): first clockwise (CW) prior to ~12 Ma and then counterclockwise (CCW) from ~12 to 5.5 Ma. The triangle, star and square indicate declinations of the average interval–mean direction (357.7±3°), the present geomagnetic field (~3°) and the expected direction for Eurasia at 10 Ma (8.9±3.6°), respectively.

[63–65] and one in the Turfan Basin [34] lies at 72.9° N, 246.2° E (K=117.7 and A_{95}=8.5°). Since the Tian Shan intramontane and foreland basins to the east of the Talas–Ferghana fault seem not to have been subjected to detectable neotectonic rotation relative to Eurasia during these times [34,63–65], the above CCW rotation of the Kuche Depression relative to Eurasia and Tarim defines rotation limited to a relatively small tectonic block.

A plot of interval–mean declinations as a function of age (Fig. 9) shows that the cumulative CCW rotation actually comprises two long-term phases of rotation namely an early clockwise (CW) rotation at a rate of ~1.36°/Ma between ~30 and ~12 Ma and a subsequent CCW rotation at a rate of ~1.64°/Ma from ~12 Ma until ~5.5 Ma. Although our record ceases at this time, we have no reason to believe that the younger CCW rotation has not continued to the present day (Fig. 9) and it may explain the CCW rotation of the Yaha section [29] which, as noted above, we believe to be mostly younger than the succession studied here. The inferred CW-to-CCW rotation transition took place some 3 Ma after major change in the regional stress regime influenced uplift and sedimentation and there is also no apparent rotational expression of the second change in provenance at ~7 Ma. The regional rotations are therefore not directly coupled to the orogenic thrusting, crustal loading and flexure and are possibly triggered by interactions with adjoining crustal blocks.

7. Conclusions

The uplift and deformational history of the Chinese Tian Shan Range deduced from magnetostratigraphic study of a Paleogene and Neogene composite section in the Kuche Depression can be summarized under four headings:

(1) The changes in rock magnetic susceptibility parameters (k_m, P_j, T) are used as proxies for changes in sediment provenance accompanying changes in the regional tectonic regime. The magnetostratigraphic correlation proposed in this paper implies that the southern Tian Shan Range was reactivated at ~20 Ma with initiation of uplift presumably induced by tectonic stress imparted during the interval ~20–15 Ma and the resulting in uplift causing magnetite-rich detritus to be shed into the depression developed by flexural loading.

(2) Rapid uplift of the southern Chinese Tian Shan appears to have occurred at ~16–17 Ma, a few million years after initiation of thrusting in the Kuche Depression. This pulse of rapid uplift nearly doubled sediment accumulation rates and is considered to have been dominantly controlled by tectonism with climatic changes playing little or no part (e.g. [15,16,18]). This mid-Miocene rapid uplift accords with age estimates from balancing measured stratigraphic sections in the Tarim and Junggar basins [22] and from balanced cross sections derived from the Chinese Tian Shan Range [20]. Given that the northern Tian Shan was apparently being uplifted at ~11 Ma [17,24–26], this indicates that deformation migrated from south to north across the Tian Shan Range.

(3) Increase in sediment accumulation rate occurred in the Kuche Depression at ~7 Ma and this is interpreted to reflect another pulse of rapid uplift of the southern Tian Shan; whilst climatic changes could have modulated the sedimentary record, they do not appear to be a dominant control on sediment accumulation in western China between ~7 and ~2.58 Ma [15]. Unlike the earlier episode, this phase correlates with uplift identified in the northern Tian Shan [15] and was evidently the cause of a depositional episode at ~5–6 Ma observed within the Tarim and Junggar basins [22]. The youngest part of the composite section studied here is ~5.5 Ma in age and our data are unable to resolve the relative contributions of tectonic and climatic processes to Pliocene and Quaternary sedimentation [15,59–61]; more data from young successions over a wider region of the Tian Shan Range are required to resolve this question.

(4) Analysis of the 66 stratigraphically grouped interval–mean directions suggests a small net CCW rotation of ~10° of the Kuche Depression relative to Eurasia during the Cenozoic. However, stratigraphically-ordered declinations of interval–mean directions imply that this slight net CCW postcollisional rotation of the Kuche Depression relative to Eurasia actually consists of a long-term CW rotation at an average rate of ~1.36°/Ma prior to ~12 Ma, followed by a CCW rotation at an average rate of ~1.64°/Ma which may be continuing to the present day. These rotational regimes are temporally independent of regional flexural loading and basin formation and are therefore possibly a signature of changes in regional plate geometries; they appear to apply to a relatively small block because intramontane and foreland basins to the Tian Shan east of the Talas–Ferghana fault have not been subjected to detectable neotectonic rotation relative to Eurasia during these times [34].

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